Chankyo Peter Kim

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Ann Arbor, MI · Los Altos, CA — *Seeking a 2027 research internship: geometry & 3D vision for world models*

Research Summary

Robotics Ph.D. candidate at Michigan; research scientist intern on TRI's World Foundation Model team. I build world models that **remember, predict, and plan in 3D**, and evaluate them closed-loop. *Remember:* a persistent, renderable **3D Gaussian memory** — a structured latent state a video generator conditions on and exports as a scene (AnyWorld). *Predict:* action-conditioned **latent** rollouts that forecast a *distribution* over futures, and a **Gaussian world model** whose dynamics are rigid-body pose updates on the splats. *Plan and evaluate:* RL fine-tuning of driving policies inside the model, VLA benchmarks inside generated scenes (real2sim2real), MPPI planning through the Gaussian world model — with closed-loop benchmarks that expose where offline metrics mislead. Underneath, **equivariant architectures on Lie algebras** (ICML 2026) make these representations consistent across viewpoint and motion by construction.

Education

- **University of Michigan, Ann Arbor** — *2024 – Present (expected May 2028)*
  Ph.D. in Robotics — Advisor: Prof. Maani Ghaffari — Ann Arbor, MI, USA
  - Fellowships: SK hynix Global Researcher Scholarship (2026); Qualcomm Innovation Fellowship Finalist (2025); Daishin AI Fellowship (2024)
- **Seoul National University** — *2022*
  B.S. in Mechanical and Aerospace Engineering — Summa Cum Laude — Seoul, KR
  - GPA 4.06/4.3 (top 3 in department); military service, Republic of Korea (2017 – 2019)

Research Experience

- **Toyota Research Institute (TRI)** — *May 2026 – Nov. 2026*
  Research Scientist Intern, World Foundation Model Team — Los Altos, CA, USA
  - **AnyWorld — a video world model with a persistent, renderable 3D memory** *(lead author; under review)*
    - Replaced the sparse 3D point-cache conditioning of NVIDIA's Lyra-2 video-diffusion world model with a **feed-forward 3D Gaussian Splatting memory**: per-frame metric depth (frozen Depth Anything 3, DA3) passes through a trainable per-pixel Gaussian head and accumulates into one renderable, exportable 3DGS field — the model's memory *is* a scene, queryable from any viewpoint.
    - **Persistence closes the coverage gap:** on Waymo drives the memory keeps **100% novel-view coverage** where single-frame reconstruction leaves up to 23% of the view unobserved, gaining up to **+4.2 dB PSNR** (rendered memory vs. GT frame) at 45 m of ego displacement — the gap widens with distance.
    - Rendering that memory as conditioning and fine-tuning only a ∼0.3B-parameter conditioning adapter (backbone frozen; FSDP on up to 16 GPUs) made generation **2.2× less flickery** (adjacent-frame LPIPS) and sharper on held-out drives, degrade gracefully as ego speed leaves the training band (**+3 dB PSNR** over the base model, gap widening with speed), and **obey the commanded trajectory speed** (1.05–1.1× real ego motion vs. 3–4× for the equally trajectory-conditioned base model, reproducing a real stop). Now training the Gaussian head **jointly through the diffusion loss**, so memory and generator are learned as one system.
    - Built the evaluation around it: coverage-vs-ego-displacement, flicker, and optical-flow speed-adherence metrics, GT-referenced on real ego-paths with seeds and paired tests; retired a sharpness proxy that rewarded collapse and documented the failure modes (out-of-band speed silently emptying the 3D cache, checkpoint rankings inverting with speed). Weights and scripts released internally with versioned manifests.
  - **Prediction-target study — which latent should a world model predict in?** Compared frozen video encoders against image encoders as latent prediction targets for action-conditioned dynamics: quantified when **dynamics-aware video features** win and showed, probing frozen encoders as addressable memory, that video-native features **re-identify occluded objects** through token-relational structure; informed the team's backbone choice.
  - **Real2Sim2Real — generated 3D scenes as policy-evaluation environments (AnyWorld × PolaRiS)** *(Isaac Sim; ongoing)*
    - From a **single frame** rendered off a PolaRiS real-to-sim scan, AnyWorld generates the scene; DA3 depth over the generated frames accumulates into a 4.6M-Gaussian field that is decimated and converted into PolaRiS's scene format — a **generated background replaces the photographed scan** while robot, objects, camera path, and initial conditions stay byte-identical (exactly one file differs).

  - Ran PolaRiS's food-bussing benchmark **end-to-end on the generated background**: five VLA policies (two π_0 checkpoints, $\pi_{0.5}$, π_0-FAST, a PaliGemma-based policy) $\times$ 3 backgrounds $\times$ 15 rollouts. Grasp and placement rates are comparable to the scanned scene (**lift 37% vs. 27%, bowl placement 15% vs. 12%**) and the **policy ranking matches** ($r = 0.87$, $n = 5$). The control tells the harder truth: a $4\times$-degraded scan matches too ($r = 0.975$) because the background is 96% of pixels but 0% of dynamics — motivating the dynamics-bearing scenes now in progress. Closed the loop **real2sim2real** on TRI's physical DROID station: the ranking measured in the generated scene holds on the real robot.
- **Closed-loop RL fine-tuning of end-to-end driving policies inside the world model** *(patent filed)*
  - Designed an **action-conditioned latent world model** that, given a candidate action sequence, forecasts a *distribution* over future latent states rather than a point estimate — counterfactual rollouts and their uncertainty as first-class outputs *(paper in preparation)*.
  - Built the framework that uses it as a **learned simulator** to fine-tune a diffusion action head for end-to-end driving with closed-loop RL, scored on held-out scenarios instead of non-transferring open-loop metrics; adopted by TRI's end-to-end driving team as its **shared closed-loop training and evaluation stack**, passed a **Toyota Motor Corporation technical review**, and entered the TMC production pipeline with additional funding and compute.

- **University of Michigan** *Sep. 2023 – Present*
  Graduate Student Researcher — Spatial AI & World Models (Prof. Maani Ghaffari) Ann Arbor, MI, USA
  - **Gaussian world models for manipulation** (GSWM; with Prof. Xingxing Zuo's group, MBZUAI)**:** an action-conditioned world model over a 3DGS scene whose Gaussians ride on rigid bodies, so dynamics collapses to per-body pose updates and the renderer never retrains — cheap enough for a sampling planner's inner loop. MPPI planning through it took a 24-task push-to-goal benchmark from 0/24 (first model, uncapped actions) to **49% per attempt** (79% best-of-3) — see *Pushing Gaussians* below.
  - **Equivariant 3D vision:** E3DGS — proved photometry (spherical harmonics up to degree 2) is algebraically isomorphic to a rank-2 geometric tensor, giving a single $SE(3)$-equivariant treatment of geometry *and* color in 3D Gaussian Splatting; ReLN-Gaussian-MAE — a $GL(3)$-equivariant self-supervised backbone for Gaussian splats: **95.2%** accuracy under arbitrary rotations vs. 18.3% for non-equivariant baselines (ShapeSplat).
  - **Geometric deep learning:** Reductive Lie Neurons — $SO(3)/GL(n)$-equivariant architectures on Lie algebras (**ICML 2026**, [C1]).
  - **Latent world models from video (self-supervised):** action-conditioned latent dynamics learned from unlabeled video; made CLIP/DINOv2 features motion-aware and embedded VLM features into Lie-algebraic latents for open-vocabulary scene rollouts.
  - **Pose estimation / SLAM:** data- and parameter-efficient equivariant neural inertial odometry (GINIO) for GPS-denied navigation, deployed in real time on legged robots via sim-to-real *(patent filed)*.

- **Bear Robotics** *Jan. 2022 – Apr. 2022*
  Research Intern, Robotic Automation Team Redwood City, CA, USA
  - Developed **factor-graph** optimization for **visual-odometry SLAM** (camera motion estimation, landmark constraints) and improved robustness under real-world noise and outliers.

- **OUTTA [Homepage]** *Jan. 2021 – Present*
  Co-founder Seoul, KR
  - Co-founded an AI/Robotics education non-profit; AI bootcamps for 3,000+ underprivileged students and early-career developers.

SELECTED PUBLICATIONS / C: CONFERENCE, J: JOURNAL, W: WORKSHOP

[C4] AnyWorld: What Should a World Model Remember? Coverage over Fidelity in 3D Memory
Kim, C., Guizilini, V. C., Ghaffari, M. *CVPR, 2027 (in submission).*

[C3] GINIO: General Equivariant Interface for Neural Inertial Odometry
Kim, C., Lin, T.-Y., Zhu, M., Liu, B., Ghaffari, M. *CoRL, 2026 (under review).*

[C2] E3DGS: Unified Geometric–Photometric Equivariance for 3D Gaussian Splatting via Color-as-Geometry Embedding
Kim, C., Ghaffari, M. *ICLR, 2027 (under review).* [arXiv]

[C1] Equivariant Neural Networks for General Linear Symmetries on Lie Algebras (Reductive Lie Neurons)
Kim, C., Zhao, S., Zhu, M., Lin, T.-Y., Ghaffari, M. *ICML, 2026.* [arXiv] [project page]

[J1] SO(3)-Equivariant Neural Inertial Odometry
Kim, C., Lin, T.-Y., Zhu, M., Liu, B., Ghaffari, M. *IEEE RA-L, 2026 (under review).*

[W2] Reductive Lie Neurons: A Framework for $\mathrm{GL}(n, \mathbb{R})$-Equivariant Deep Learning
Kim, C., Zhu, M., Ghaffari, M. *RSS Workshop on Equivariant Systems, 2025.* [link]

[W1] Equivariant Neural Inertial Odometry
Kim, C., Zhu, M., Ghaffari, M. *ICRA Workshop, 2025.* [link]

Patents: two filed — world-model-in-the-loop training and closed-loop evaluation of driving policies (TRI, 2026); equivariant neural inertial odometry (Univ. of Michigan).

Selected Research Projects — 3D Vision & World Models

- **Pushing Gaussians — planning through a learned world model over a 3DGS scene** *2026 – Present*
  Gaussian world model · MPPI×CEM planner (192 samples, horizon 60) · 24-task push-to-goal benchmark UMich × MBZUAI
  - Closed $0 \rightarrow 49\%$ with four measured levers (cap actions to the training distribution — an optimizer finds where the model is wrong; train rollout length to cover the planning horizon; execution budget; retry) against eleven hypotheses tested and rejected ($\geq$2 repeats each). Model error is 5.9 mm at 8 steps but 33 mm at the 60-step horizon (tolerance 30 mm): every lever that worked kept the planner inside the model's trust region.
  - **Finding that transfers:** offline metrics (one-step, 40-step rollout error) did not predict — at times anti-correlated with — closed-loop success; the checkpoint with the best rollout error planned at a third of the champion's rate. **A planning world model must be selected in closed loop.**
- **E3DGS: Unified Geometric–Photometric Equivariance for 3D Gaussian Splatting** *2026*
  Color-as-geometry embedding; $SE(3)$-equivariant processing of Gaussian splats without tensor products [arXiv]
  - Showed the Wigner-D action on spherical-harmonic color reduces to matrix conjugation, so every Gaussian attribute embeds in one $\mathfrak{gl}(3)$ space; better robustness to camera transformations and data efficiency on object vision and action-conditioned scene modeling.
- **GINIO / Equivariant Neural Inertial Odometry** *2024 – 2026*
  $SO(3)$-equivariant learned odometry from IMU alone; a general equivariant interface for existing NIO backbones Univ. of Michigan
  - Data- and parameter-efficient pose estimation for GPS-denied navigation; deployed in real time on legged robots via sim-to-real *(patent filed; CoRL and RA-L submissions under review)*.

Grants and Honors

- **Fellowships:** SK hynix Global Researcher Scholarship (2026 –); Qualcomm Innovation Fellowship Finalist (2025); Daishin Graduate AI Research Fellowship (2024).
- **Scholarships:** Sinyang Foundation (2020 – 2022); SNU Academic Excellence, 4 semesters (2015 – 2016).
- **Awards:** Summa Cum Laude, SNU (2022); Samsung Humantech Paper Award, Bronze (2014); Korea Physics Olympiad, Silver (2011).

Academic Activity & Service

- **Selected Participant** — Machine Learning Summer School (MLSS), Columbia University *2026*
- **Invited Seminar** — "Why Equivariance? Foundation Models in Robotics", SNU *Dec. 2025*
- **Talk** — "Toward Equivariance on Uncertainty and Reductive Lie Neuron Design", CVPR 2025 *June 2025*
- **Reviewer** — ICML, CoRL (2026); IEEE TRO, ICRA, IROS (2024–2025)
- **Research Mentorship** — 3 MS students: world modeling, neural VIO/SLAM, inertial odometry *2025 –*
- **Teaching (GSI)** — Computational Symmetry in AI & Robotics; Mobile Robotics; Self-Driving Cars *2024–25*

Technical Skills

- **3D Vision & Geometry:** 3D Gaussian Splatting (gsplat), feed-forward reconstruction, metric depth (Depth Anything 3), SfM/SLAM, factor graphs, IMU–visual fusion, equivariant networks on Lie groups.
- **World Models & Generation:** video-diffusion world models (Lyra-2, Wan, Cosmos), action-conditioned latent dynamics, fine-tuning frozen priors, world-model-in-the-loop RL, VLA policies (π_0 family).
- **Planning, Evaluation & Simulation:** MPPI/CEM planning in learned models, closed-loop policy evaluation (PolaRiS real2sim2real), world-model benchmarks (coverage, stability, speed adherence), Isaac Sim, Habitat.
- **Systems & Languages:** PyTorch, JAX; multi-GPU FSDP training (up to 16 GPUs); HuggingFace releases; Python, C++.

Books & Educational Materials

Author of four Korean textbooks on AI, robotics, and Python: *NLP and Data Analysis* (Hongreung, 2023) [link]; *Python and Autonomous Robot Systems with MIT Race Car* (Hongreung, 2022) [link]; *Calculus and AI with Advanced Python* (Hongreung, 2022) [link]; *Life as an Engineering Student* (MegaStudy, 2021) [link].